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A review of mineral precipitation and effective scale inhibition methods at geothermal power plants in West Anatolia (Turkey)

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ABSTRACT

In water-dominated reservoirs, binary and flash cycle geothermal power systems are widely used for power generation at moderate or moderate-high reservoir temperatures. For effective management of both energy systems, it is critical to understand the effects of controlling pressure in different sections of a power plant as well as temperature changes and the pH levels of geothermal fluids. For advanced (multi-flash or flash + binary systems) systems, although the expectation of energy efficiency is higher than for flash and binary systems, the management of these systems is more complex because of high reservoir temperatures and multi-separation processes. These multi-steam separations result in dramatic pressure and temperature drops for geothermal fluids after each separation stage.

Depending on the geothermal reservoir rocks, water-rock interaction can be inferred and possible scale types may be predicted through water-modeling programs. Based on this information, suitable inhibitors may be suggested after short-term field tests prior to start up processes for geothermal power plants.

Western Anatolia has important high-medium geothermal systems that are suitable for power production, depending on large graben structures in the Aegean Extensional Zone. Geothermal power plants (Turkey) are generally designed as binary and flash types, based on the reservoir temperatures (in Germencik-Aydın, Kızıldere-Denizli, Alaşehir-Manisa regions etc.) in Western Anatolia. The scale types may change with respect to geothermal fields even if they are in close proximity to each other. At the same time, in addition to specific scale types, some minerals are commonly found across each of the geothermal fields. The most important scale types in production wells and lines as well as surface equipment are carbonate minerals such as calcite and aragonite. In addition, silica minerals also tend to form scale, up to 150 ^{o}C. Scale types in reinjection lines and wells have been commonly identified as celestine, strontium and barite minerals mixed with silica and carbonates in some geothermal power plants. But these scale types also differ within the same reservoir, with well depth. Scale prevention inhibitors and inhibitor systems with different chemical properties are generally used for anti-scaling. Chemical inhibitor performance and mineral precipitation can be monitored both periodically chemical analysis and also using steel control coupons, which are put in at critical pressure drop points along pipelines.

Controlling re-injection temperatures and pH levels have had success for anti-scaling. This study is mainly focused on mineral precipitation conditions and the most effective scale inhibition applications in different type geothermal power plants in Western Anatolia.

1. Introduction

Among all renewable energy sources, geothermal energy is a suitable option for heating and power production. It is both a sustainable source and independent of meteorological conditions. Geothermal power production has higher capacity factors (> 80%) compared with other renewable energy resources and it has an advantage in the capability of supplying base load electricity for the investors (IRENA, 2017). The total installed capacity of geothermal power plants globally in 2018 reached 15 GW h, and this capacity is expected to rise to 17 GW by 2021, with the largest capacity additions expected in Turkey, Indonesia, the Philippines and Mexico (IAEA, 2017a; Fig. 1).

In addition to the high capacity factor of geothermal power plants, they assume an important role, similar to other renewable energy sources such as such as wind, solar, hydro, in reducing CO_2 emissions globally. Global carbon emissions due to the burning of fossil fuels have

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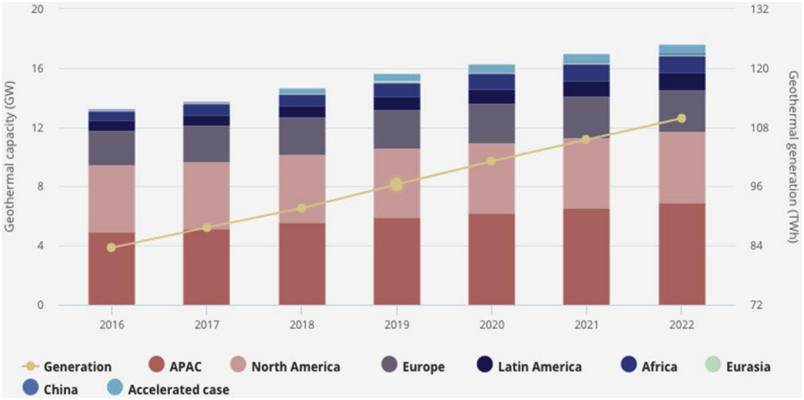


Fig. 1. Geothermal power generation and cumulative capacity by region, 2016–2022 (IAEA, 2017a).

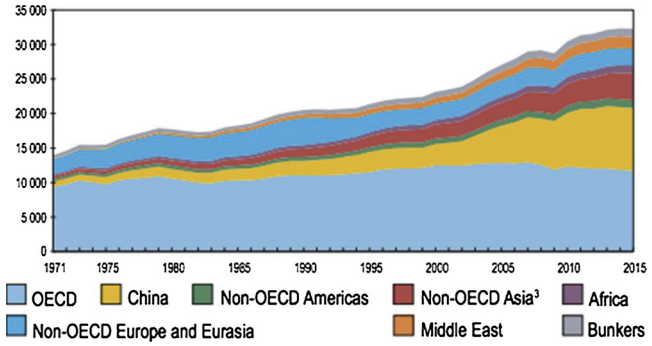


Fig. 2. World CO_2 emissions from fuel combustion from 1971 to 2015 by region (Mt of CO_2 IAEA, 2017b).

significantly increased since 1900 (Fig. 2). From 1970 to 2017, CO_2 emissions increased by approximately 90% from fossil fuel combustion and industrial processes, contributing to about 78% of total greenhouse gas emissions. Atmospheric CO_2 concentrations exceeded 410 ppm in May 2018, according to The Mauna Loa Observatory, which is an atmospheric baseline station in Hawaii. Based on the Kyoto Protocol (2005) and Paris Climate Agreement (2015), CO_2 emissions must be decreased, and this can be accomplished primarily through greater reliance on renewable energy sources, and less on fossil fuels.

Among countries with geothermal energy production, Turkey is located in the top four, with 41 numbers geothermal power plants around the world in 2018. Total installed geothermal power capacity of the country is approximately 1.4 GWe for 2018 (EMRA, 2018; Fig. 3) and the new target set by the Turkish Ministry of Energy and Natural Resources is 4000 MWe by 2030.

Although geothermal power is a green option, with tremendous

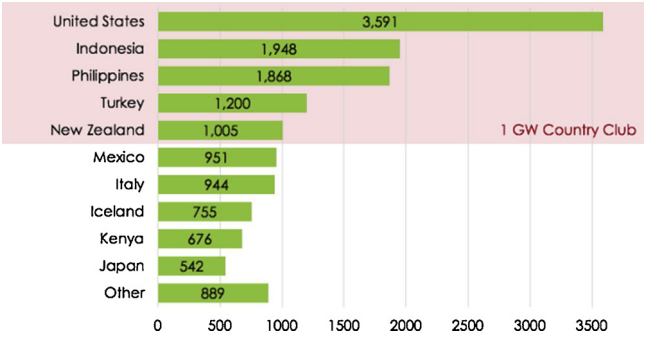


Fig. 3. Installed geothermal power production around the world. Turkey will reach 1.4 GWe of installed geothermal power capacity by the end of the 2018 (by www.thinkgeoenergy.com).

capacity factors in relation to other renewables, the operational period is critical in order to achieve the installed capacity of a power plant. Thermodynamic changes based on pressure and temperature from depth to surface conditions, non-condensable gases (NCG) as well as issues with steam quality may cause serious problems for power plant operations. Common problems are scales in boreholes and surface equipment (e.g., pipes, separator systems, vanes, pumps, steam turbine blades, cooling towers), corrosion due to NCGs such as CO_2 and H_2S, and low pH conditions in both water dominated and steam dominated reservoirs (Haklıdır Tut and Haklıdır, 2017). The provision of continuous steam production and power generation depends directly on the monitoring of the geothermal system and addressing these problems with different treatment methods. Geothermal power generation can be provided from organic rankine (ORC) type binary cycle, flash type, or advanced geothermal systems (flash + binary), depending on reservoir temperatures in water dominated systems (DiPippo, 2016). Operational problems can vary, depending on the type of power cycle. For example, operation and maintenance issues such as geothermal scale inhibition, corrosion prevention, and cooling tower treatment for multi-flash or advanced power systems are more complex than for binary systems (Haklıdır and Şengün, 2016).

ORC-type geothermal power plants have been installed worldwide (Bertani, 2015). Preference for this type of plant is driven by ease of installation and maintenance, as well as the requirement of lower reservoir temperatures. With this prevalence, researchers have more experience with ORC type geothermal power plants around the world. However, geothermal fluids expose complex thermodynamic changes at multi-flash and advanced systems to, requiring the systems to have a serious attention during operational periods. In Turkey, the growth of the geothermal power industry has accelerated since 2009. There are a few installed flash and advanced type power plants in the country, and the rest of them are ORC type binary cycle, all located in Western Anatolia. While studies have been carried out on geothermal scaling and scale inhibition on ORC binary cycles in Western Anatolia (e.g. Demir et al., 2014; Baba et al., 2015; Çelik et al., 2017; Yıldırım and Yıldırım, 2015), research is limited on operational challenges such as geothermal scale precipitation for flash and advanced type power plants, (Osborn et al., 2007; Haizlip Robinson and Haklıdır Tut, 2011; Haklıdır and Şengün, 2016; Haklıdır Tut and Haklıdır, 2017) and there are a few papers on geothermal scaling and scale inhibition on ORC binary cycles such as; Demir et al., 2014; Baba et al., 2015; Çelik et al., 2017; Yıldırım and Yıldırım, 2015 in Western Anatolia.

2. High temperature geothermal systems and power production in Western Anatolia (WA)

Since the 1960s, more than 200 geothermal fields have been identified in Turkey by the General Directorate of Mineral Research Application (MTA), and geothermal exploration continues. Nearly all

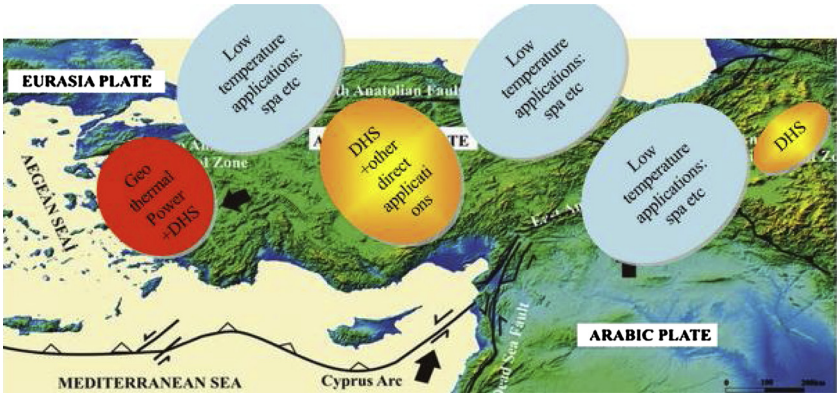


Fig. 4. Existing Geothermal Energy Applications in Turkey (Haklıdır Tut, 2015).

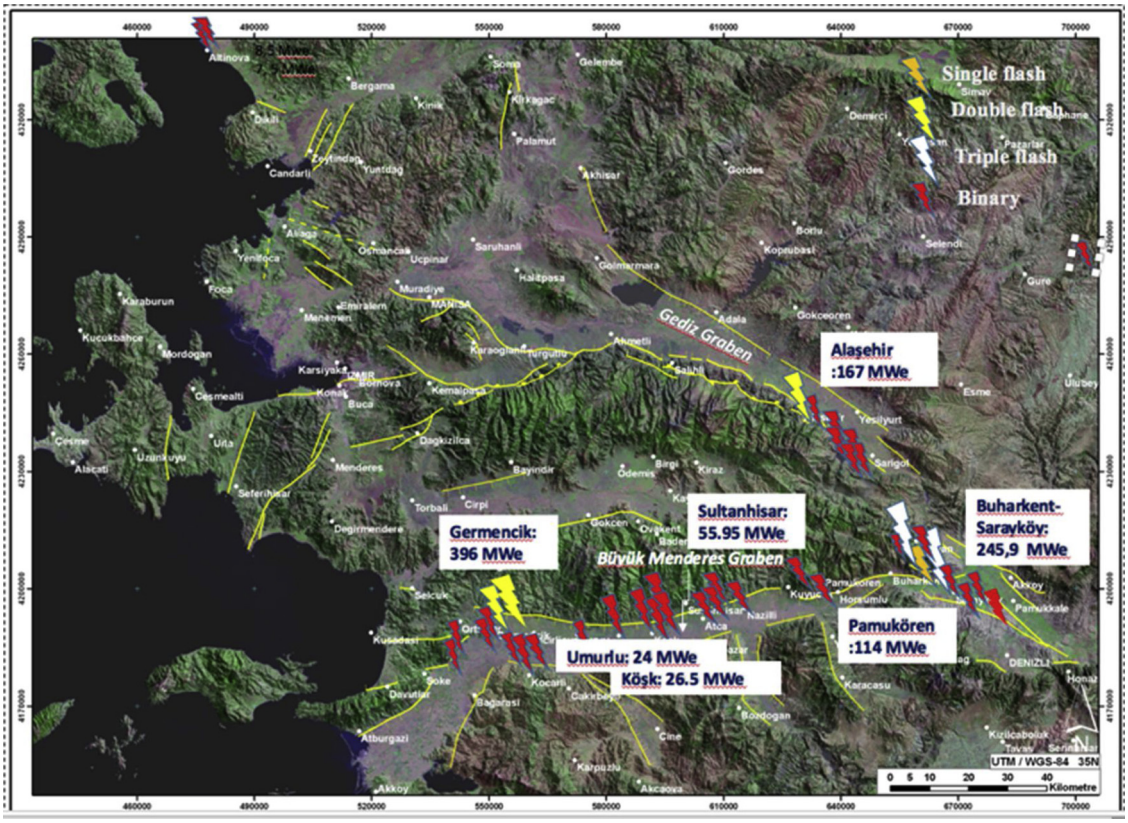


Fig. 5. Installed geothermal power plants in Turkey (2018).

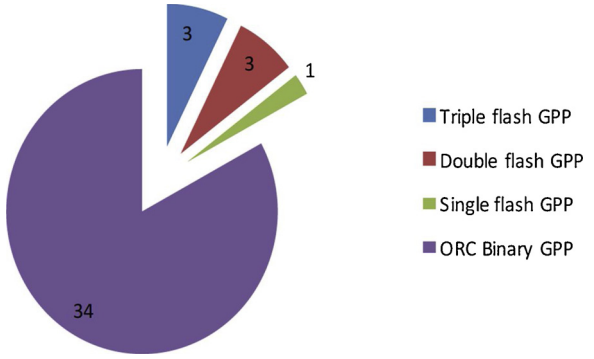


Fig. 6. Number of geothermal power plants, by type, in Turkey (2018).

identified geothermal systems have water-dominated reservoirs and temperatures are quite different from the West to the East. Western Anatolia has large graben systems and the highest reservoir temperatures (up to 245 ^{0}C), while the middle and the eastern regions have lower reservoir temperatures. Based on the reservoir temperatures in these parts, Western Anatolia has been deemed more suitable for geothermal power production (Fig. 4) while the other regions have used this resource primarily for district heating and low temperature applications (Haklıdır Tut, 2015). For 2018, Turkey's installed geothermal power capacity is approximately 1100 Mwe and the installed heating capacity is around 3300 Mwt (TJD, 2018).

Turkey is located on the Alpine-Himalayan orogenic belt and it is affected by different tectonic zones such as the North Anatolian Fault, the Eastern Anatolian Fault, and the Aegean Extension Zone. Some volcanics have been identified in the Western region of the country (Kula volcanics) as well as the Eastern region (Nemrut volcanics) (Bozkurt, 2001; Haklıdır Tut, 2007). These geological activities provide

Table 1
Characteristics of thermal waters in selected geothermal fields in West Anatolia.

No	Well	Type	Location	$T_{res.}$	pH	EC (μS/cm)	Water Type	References
1	KD-20	Production	Kızıldere	204	8.4	4220	$Na\text{-}HCO_3$	Tarcan et al., 2016
2	KD-21	Production	Kızıldere	205	8.4	4190	$Na\text{-}HCO_3$	Tarcan et al., 2016
3	KD-22	Production	Kızıldere	204	8.1	4120	$Na\text{-}HCO_3$	Tarcan et al., 2016
4	R-1	Production	Kızıldere	242	8.1	5480	$Na\text{-}HCO_3$	Tarcan et al., 2016
5	R-3	Production	Kızıldere	204	9.3	5320	$Na\text{-}HCO_3$	Kındap and Haklıdır Tut, 2011
6	K38A	Reenjection	Kızıldere	180	8.2	5500	$Na\text{-}HCO_3$	Zorlu,2014
7	KD-18A	Production	Kızıldere	230	7	5040	$Na\text{-}HCO_3.SO_4$	Haklıdır and Şengün, 2016
8	KD-23B	Production	Kızıldere	243	6.7	5320	$Na\text{-}HCO_3.SO_4$	Haklıdır and Şengün, 2016
9	KD-23D	Production	Kızıldere	238	6.9	5500	$Na\text{-}HCO_3.SO_4$	Haklıdır and Şengün, 2016
10	A-1	Production	Alaşehir	190	6.8	2630	$Na\text{-}HCO_3$	Haklıdır and Şengün, 2016
11	AG-22	Production	Germencik	195	8.5	7330	$Na\text{-}Cl\text{-}HCO_3$	Tarcan et al., 2012
12	B-10	Production	Germencik	230	7.98	7130	$Na\text{-}Cl\text{-}HCO_3$	Tarcan et al., 2012
13	BR-1	Reenjection	Germencik	171	7.12	4440	$Na\text{-}Cl\text{-}HCO_3$	Tarcan et al., 2012
14	BR-2	Reenjection	Germencik	169	6.99	4480	$Na\text{-}Cl\text{-}HCO_3$	Tarcan et al., 2012
15	T-9	Production	Tuzla	149.1	6.14	90.60	NaCl	Demir et al., 2014
16	T-9 Separator	Production	Tuzla	142	7.74	93.20	NaCl	Demir et al., 2014
17	T-16	Production	Tuzla	150.6	6.20	91.40	NaCl	Demir et al., 2014
18	T-16 Separator	Production	Tuzla	142.5	7.60	91.90	NaCl	Demir et al., 2014
19	S-1	Production	Salavatlı	170	7.12	4440	$Na\text{-}Cl\text{-}HCO_3$	Tarcan et al., 2012
20	SR-2	Reenjection	Salavatlı	169	6.99	4480	$Na\text{-}Cl\text{-}HCO_3$	Tarcan et al., 2012
21	SR-4	Reenjection	Salavatlı	170	7.6	4530	$Na\text{-}Cl\text{-}HCO_3$	Tarcan et al., 2012
22	S-2	Production	Salavatlı	172	7.16	4460	$Na\text{-}Cl\text{-}HCO_3$	Tarcan et al., 2012
23	S-3	Production	Salavatlı	170	7.33	4530	$Na\text{-}Cl\text{-}HCO_3$	Tarcan et al., 2012
24	S-4	Production	Salavatlı	174	7.48	4560	$Na\text{-}Cl\text{-}HCO_3$	Tarcan et al., 2012
25	HB-7	Production	Germencik	170	8.78	6210	$Na\text{-}Cl\text{-}HCO_3$	Tarcan et al., 2012
26	HB-5	Production	Germencik	174	8.28	5930	$Na\text{-}Cl\text{-}HCO_3$	Tarcan et al., 2012
27	RE-1	Reenjection	Germencik	154	8.35	5450	$Na\text{-}Cl\text{-}HCO_3$	Tarcan et al., 2012
28	RE-2	Reenjection	Germencik	120	7.21	5720	$Na\text{-}Cl\text{-}HCO_3$	Tarcan et al., 2012
29	HB-8	Production	Germencik	168	6.22	5730	$Na\text{-}Cl\text{-}HCO_3$	Tarcan et al., 2012
30	AP-2	Production	Pamukören	182	8.5	5100	$Na\text{-}HCO_3$	Karahan et al., 2015
31	AP-3	Production	Pamukören	183	8.71	5000	$Na\text{-}HCO_3$	Karahan et al., 2015
32	AP-7	Production	Pamukören	191	8.86	4970	$Na\text{-}HCO_3$	Karahan et al., 2015
33	AP-13R	Reenjection	Pamukören	159	8.69	4960	$Na\text{-}HCO_3$	Karahan et al., 2015
34	AP-14R	Reenjection	Pamukören	177	8.87	4840	$Na\text{-}HCO_3$	Karahan et al., 2015

access to geothermal sources across Turkey with a considerable range of temperatures (Haklıdır Tut, 2015).

In Western Anatolia, there are a few large graben systems and two of them have substantial geothermal energy potential, such as the Büyük Menderes Graben (BMG) and the Gediz Graben (GG), with high-medium reservoir temperatures that are quite suitable for power production (Fig. 5).

The BMG geothermal systems have higher reservoir temperatures than the GG and these are the towns of Germencik (234 °C), Umurlu, Köşk, Sultanhisar (170 °C), Pamukören (190–200 °C), which are connected to the city of Aydın and the town of Sarayköy (245 °C; Denizli city) (Haklıdır Tut, 2017). The southern section of the GG is also suitable for power production; five geothermal power plants are located in the town of Alaşehir (Manisa) (Fig. 5).

Most of the existing geothermal power plants utilize an ORC binary cycle, with a few using single and multi-flash systems. In the Germencik region (47.4 × 2 MWe_{gross}) in the western section of the BMG, and in the Alaşehir region (30 + 15 ORC MWe_{gross}) of the GG, there are double flash systems. In the eastern section of the BMG, the Kızıldere region has one single flash (Kızıldere I 17.2 MWe_{gross}), and two triple flash + binary systems (Kızıldere II- 60 MWe_{gross} flash type + 20 MWe_{gross} ORC Kızıldere III Unit-1: 72 MWe_{gross} flash type +23 ORC; Unit-2: 50 MWe_{gross} +23 MWe_{gross} ORC) (Fig. 6). Overall, four advanced geothermal power systems are located along the graben systems.

3. Scale precipitation mechanisms and characterics of geothermal systems in Western Anatolia

Scale precipitation is a major problem in water-dominated geothermal systems. When it takes place in a geothermal production borehole, the production rate of geothermal fluids is reduced, resulting in a gradual decrease in net energy production (Haizlip Robinson et al., 2012). As a general definition, scale is the precipitation of minerals in different thermodynamic conditions in geothermal waters. Pressure, temperature, pH and salt concentration directly affect the solubility of the minerals in these waters. From deeper zones to surface conditions, geothermal fluids are enriched by minerals due to water-rock interaction under high P, T conditions (Haklıdır Tut and Haklıdır, 2017). This means that it is important to understand geothermal reservoir rock types in order to predict possible scale precipitation in a geothermal field.

Geothermal fluids consist of hot water, non-condensable gases, and steam in water dominated systems. Hot water at depth in a geothermal system contains dissolved minerals and dissolved gases at high temperatures and high pressure. Under these dynamic reservoir conditions, when measured total pressure is lower than $P_{gas} + P_{liquid}$, the flashing process begins in deeper zones in a geothermal well (Haizlip Robinson et al., 2012).

As geothermal fluids are discharged from deep to surface conditions through a well, a change in thermodynamic conditions can result from this boiling process, which releases dissolved gases and changes the pH to a more alkaline level. This essentially leads to the precipitation of minerals such as calcite, and some metal sulfides (stibnite) and silica minerals that can deposit very hard scales inside a well as well as on the inside surface of equipment such as separator systems and steam turbine systems (Morris and Mroczek, 2015; Haklıdır Tut and Haklıdır, 2017). When the water dominated geothermal fluid is flashing at a lower separator pressure in the power plant, the separated water phase becomes oversaturated with silica minerals (particularly amorphous silica) as its solubility decreases with a temperature drop. The temperature of geothermal fluids may vary between 160 °C and 250 °C in high temperatures reservoirs. After steam separation, the brine is saturated with amorphous silica and amorphous silica starts to precipitate

Table 2
Chemical composition for the thermal waters in selected geothermal fields in West Anatolia (Numbers are the same as in Table 1).

Sample No	Na mg/l	K mg/l	Ca mg/l	Mg mg/l	HCO_3 mg/l	Cl mg/l	SO_4 mg/l	SiO_2 mg/l	B mg/l	Li mg/l	Ba mg/l	Sr mg/l
1	1246	121.1	11.6	0.5	2587	98	816	263.9	19,119	3,498	0.073	1.47
2	1282	124.8	10.9	0.5	2306	92	768	287.8	18,033	3,459	0.083	1.55
3	1291	127.7	13.2	0.6	2258	88	708	306.2	17,061	3,502	0.082	1.81
4	1534	207	3	1	2978	96	381	477.8	29,799	5,646	0.02	0.48
5	1225	104	2.48	1	1787	140	850	227	17	–	–	–
6	1001	107	1	1.23	1729	92.9	577	402	–	2.8	–	–
7	1165	220	1.45	< 1	2577	115	638	552	–	3.8	–	–
8	1324	228	1.33	< 1	2515	110	653	697	–	4.3	–	–
9	1165	220	1.08	< 1	2487	104	591	503	–	–	–	–
10	570	100	5.65	< 1	1333	174	8.8	500	99	6	–	–
11	1635	12.84	119	25.4	1008	1790	33	589	71.97	8,893	0.661	0.887
12	1485	11.03	168	34.3	1220	1707	21	518.5	67.75	6,758	0.478	0.626
13	1167	89	10.2	3	2712	105	123	211.1	52.34	4,347	0.186	0.98
14	1180	88	6.6	1	2735	103	102	206.2	53,663	4,326	0.112	0.79
15	19320	1.9	3.02	115	118	34740	16	195	60.50	21.60	5.10	145.50
16	19350	1.89	2.99	111	117	36380	181	189	60.50	21.60	5.10	124.40
17	18970	1.81	2.95	107	140	34280	167	196	75.10	25.60	5.90	175.80
18	19170	1.87	2.96	102	133	34150	165	204	64.20	22.40	5.00	148.20
19	1167	89	10.2	3	2712	105	123	211.1	52.34	4,347	0.186	0.98
20	1180	88	6.6	1	2735	103	102	206.2	53,663	4,326	0.112	0.79
21	1194	95	7.8	1	2641	102.5	78	208.5	53,703	4,917	0.125	0.99
22	1196	91	8.5	0.5	2934	110	75	231.6	51,648	4,581	0.311	0.86
23	1208	97	8.5	1	2795	94	60	237.8	54,725	5,034	0.076	1.08
24	1175	95	7.4	1	2765	110	75	222.4	55.26	4,913	0.12	0.9
25	1538	21.6	259	35.9	1196	1375	38	334.6	65.81	7,581	1,192	2,267
26	1336	16.83	191	50.4	1601	1302	32	269.1	57.89	6,307	1,273	2,859
27	1205	11.71	97.7	40.9	1070	3137	109	247.5	53.32	6,138	1,131	2,334
28	1271	96.39	270	38.5	1708	2739	31	270.4	55.22	6,271	1,077	2,287
29	1222	85.32	210	58.2	1684	2487	30	261.8	50.43	6,095	1,090	2,261
30	1272	196	3.3	0.1	2519	2519	180	336	30	–	–	–
31	1240	170	1.9	0.1	356	2306	318	356	36	–	–	–
32	1225	148	2.45	0.2	2293	335	222	395	34	–	–	–
33	1174	140	6.5	5.7	2210	296	226	312	30.5	–	–	–
34	1290	130	7.6	0.98	2160	276	238	361	29.5	–	–	–

Table 3
Non-condensable Gas Compositions of the Geothermal Systems in Western Anatolia (Haizlip Robinson et al., 2013).

Geothermal Field	Reservoir gas/steam (kg/kg)	CO_2 %	H_2S %	Ar%	N_2%	CH_4 %	H_2 %
Kızıldere Deep Reservoir	0.030	98.7	0.021	0.01	0.67	0.56	0.025
Germencik	0.021	98.5	0.21	0.001	0.44	0.70	0.035
Alaşehir	0.034	98.5	0.20	0.004	0.40	0.80	0.07

rather than the less soluble minerals quartz or chalcedony, because precipitation of the latter minerals is very slow (Gunnarsson and Arnorsson, 2005; Haklıdır Tut and Haklıdır, 2017).

In these conditions, in the absence of process intervention, the silica mineral tends to precipitate in re-injection lines and re-injection wells in the system (Haklıdır Tut, 2017).

After discharging of geothermal fluids to surface condition, temperature and pressure values start to change in borehole and some minerals tend to precipitate in fluids and mineral precipitation continues from the production phase to the re-injection phase in water dominated geothermal systems. At this point, the power cycle type is important when applying effective protection to the geothermal power system.

3.1. A brief description of the reservoir geology of the high temperature geothermal systems along BMG and GG in WA

These geothermal systems contain highly fractured zones because of the Aegean Extensional Zone effect, which involves east-west trending normal faults parallel to the graben systems and the second set of north-south trending faults in the eastern region of the BMG. The reservoir rocks at deeper zones show similarities, while the main reservoir is Paleozoic age Menderes Metamorphics for the BMG and GG.

Germencik (Aydın) (BMG) is located on the west side of the BMG. The Germencik Field is geologically incompatible with the graben's characteristic structure, consisting of Paleozoic-age Menderes Metamorphics and Cenozoic aged sediments. The Germencik field's main/deep reservoir is seen to be abundantly fractured at the fault levels of Menderes Metamorphics. Chlorite, biotite, calcite and pyrite alteration minerals are seen primarily at this level (Şimşek, 1984).

Salavatlı (Aydın) (BMG) is centrally located in the BMG. The Salavatlı field is composed of Menderes Massif metamorphic rocks and Cenozoic Sediments. The reservoir features Menderes Metamorphics, overlain by gneisses, which possess good characteristics for a cap rock. Quartz, albite, chlorite, calcite, aragonite, dolomite, illite, montmorillonite, kaolinite, gypsum, pyrite, siderite, hydro-biotite and muscovite alteration minerals are in abundance at the reservoir levels (Karamanderesi and Helvacı, 2003).

Pamukören (Aydın) (BMG) is located in the east region of the graben, and its geological sequence consists of Menderes Metamorphics and the Cenozoic age units. Menderes Metamorphics are formed in the deep reservoir rocks, as a consequence of fault zones, giving marble its melting-fractured and cracked features. The Miocene age unit, which reaches upper metamorphics and gneisses also has the reservoir property of containing fault lines (Karahan et al., 2015).

Kızıldere *(Denizli) (BMG)*, in the eastern section of the BMG,

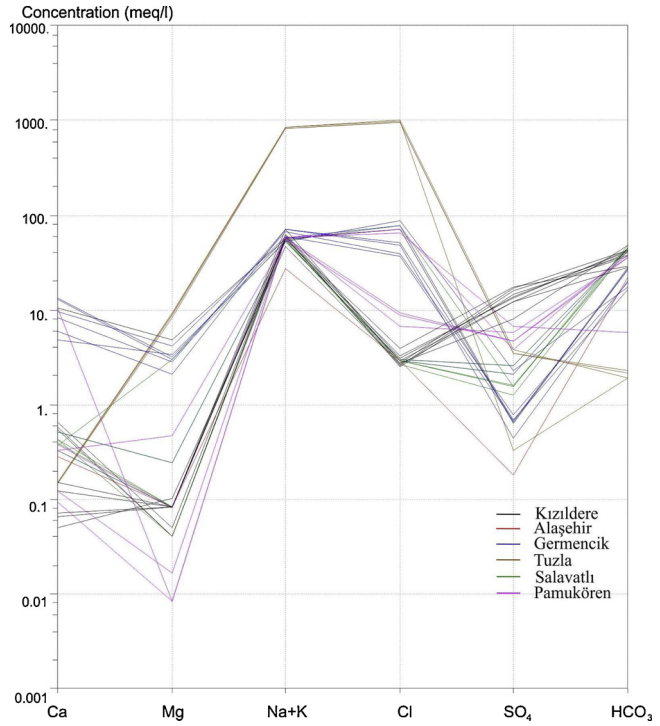


Fig. 7. The Schoeller diagram of the thermal waters for selected high temperature geothermal fields in West Anatolia.

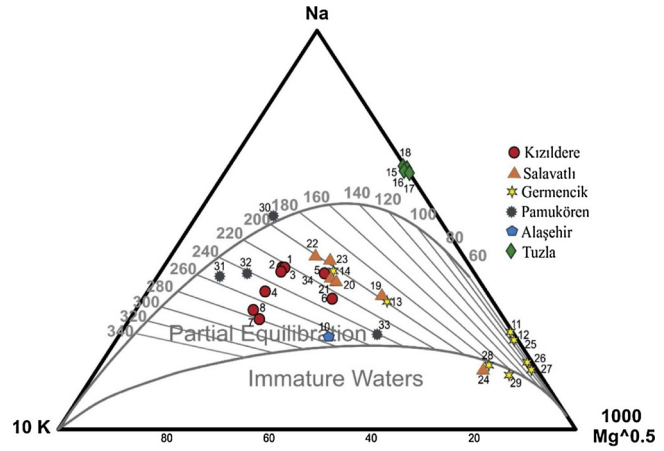


Fig. 9. Na-K-Mg diagram for the thermal waters of the selected geothermal fields in West Anatolia.

contains Paleozoic-age Menderes Metamorphics, which are the lower formation of the stratigraphic sequence, consisting of chlorite schist, mica schist, sericite schist, and quartzite. The Sazak Formation, as the first reservoir rock, is intercalated with limestone and marl, which became permeable due to tectonic moves. The İğdecik Formation is referred to as the 2nd reservoir, and a member of metamorphics, consisting of marble, mica schist, and quartzite intercalation. The 3rd reservoir rocks show the character of the Menderes Metamorphics, with deeper layers having higher temperatures, and consisting of mica schist, quartzite, calc schist, more sericite schist, and more chlorite schist (Şimşek, 2015).

Alaşehir (Manisa) (GG) is located in a basin that is divided by Menderes metamorphic rocks in the south and Quaternary sediments in the north. The base rock consists of Paleozoic aged Menderes Metamorphics, represented by high temperatures and pressures in the deep reservoir. The Mesozoic-aged Alaşehir formation overlies a metamorphic basement, represented by low temperatures in the shallow reservoir. Some alteration minerals are also seen in the sequence. An illite zone is observed in upper levels of the stratigraphic zone while an illite-chlorite zone has been identified in the reservoir layers, with sulfide mineralization also observed in patches (Bozkurt, 2001; Zorlu, 2013; Haklıdır Tut et al., 2013).

Although the main reservoir rocks consist of similar minerals, for each graben system, the chemical characterization of the geothermal

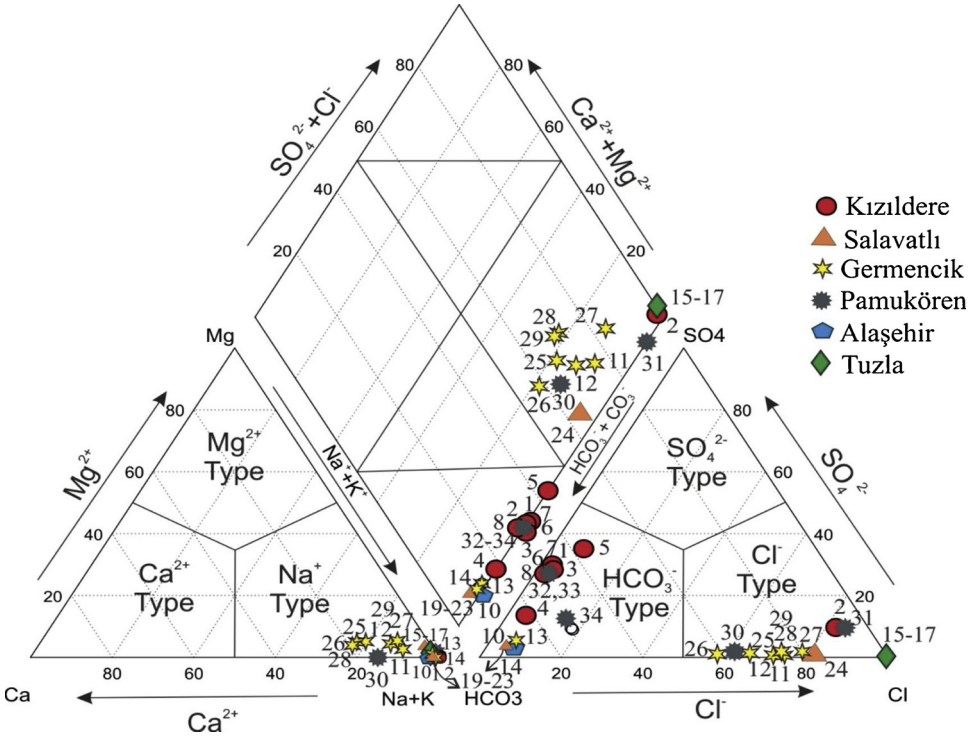


Fig. 8. Piper diagram for the thermal waters for the selected high temperature geothermal fields in West Anatolia.

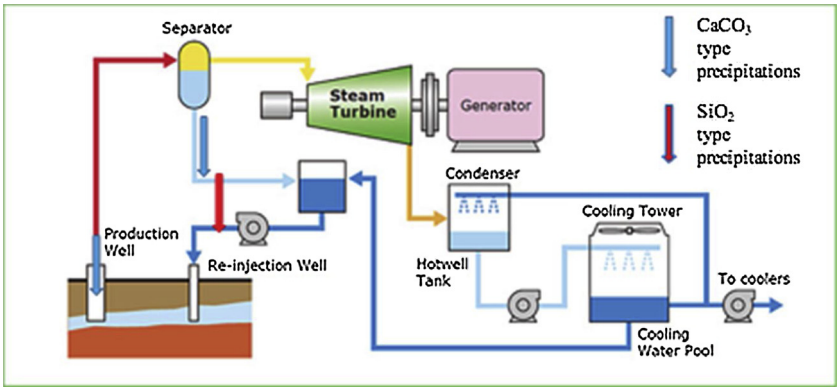


Fig. 10. Basic working principle of a single flash geothermal power plant and possible mineral precipitation points in the system.



Fig. 11. a Kızıldere-I single flash type GPP (17,2 MWe_{gross}) in the BMG. **11b.** Scale precipitation in a production line before a scale inhibition application in the Kızıldere field. **11c.** Stereoscopic enlargement of the scale sample; taken prior to a separator condition in a production well at the Kızıldere-I GPP in (Zorlu Energy, 2009).

Table 4
Scale analysis of the sample in Fig. 8c by X rays Fluorescence technique (XRF).

Element	U.M.	Value
Magnesium	% p/p	50.5
Silicon	% p/p	26.2
Calcium	% p/p	14.4
Iron	% p/p	1.4
Aluminum	% p/p	0.89
Strontium	% p/p	0.88

fluids can vary along the BMG and GG because of different effects such as sea intrusion, non-condensable gases, and other mineral effects.

Tuzla (Çanakkale) is located in the northern region of Western Anatolia. Geothermal reservoirs have been identified in the form of a shallow volcanic reservoir at a depth of between 330 m and 550 m, and a deep marble reservoir at the depth of 530 m (Demir et al., 2014). It differs from other Turkish geothermal fields in its dissolved ion concentrations in the water phase in Western Anatolia (Baba et al., 2015). The Cl^- ion concentration of geothermal fluids can reach 36,380 mg/L, which is nearly twice the concentration of seawater and is therefore a major challenge for water treatment in geothermal power plants.

3.2. Hydrogeochemical and NCG characterization of high temperature geothermal fluids along BMG and GG

3.2.1. Methodology of the study

In this study, the scale precipitation tendency of high temperature geothermal fields is investigated. Thus, the most relevant geothermal fields have been selected for the study; those that are still used for geothermal power production for different thermodynamic cycles, and have hydrochemistry data collected for different studies available. Based on the selection by these researchers, nearly all thermal water samples were collected from points after the separator. Physical parameters and chemical analyses of selected production and reinjection geothermal wells from different high temperature fields in Western Anatolia are presented in Tables 1 and 2.

The scale precipitation tendency of these geothermal well and system fluids are categorized as; flash, multi-flash/advanced system, and binary. Mineral precipitation is primarily controlled by variation of temperature, pH and pressure of fluids, and the selection of a thermodynamic cycle in geothermal power plants. In particular, identification of scale precipitation potential is a complex issue because of multi separation systems in multi-flash and advanced geothermal power plants. Understanding the scale precipitation tendency of these systems is important for identifying and applying the best scale inhibition methodology in a geothermal field. For this reason, saturation indices are calculated for each category in the study.

In this study, the PHREEQC 2.1 software program (Parkhurst and Appelo, 1999) was used to compute aqueous speciation at reservoir temperature, and fluid-mineral equilibria relying on the LLNL database. Computed saturation indices (SI) are assumed to be accurate to within ± 0.5.

3.2.2. Water and gas (in steam) phase chemistry of the geothermal fluids

The physical measurements and chemical analyses of selected thermal waters are shown for 34 selected thermal water samples in Tables 1 and 2. The reservoir temperatures of geothermal fluids vary from 168 °C to 243 °C in the BMG, while it is around 190 °C in the GG and 142–150.6 °C in the Tuzla region for production wells (Table 1).

Along the BMG, pH and EC values vary between 6.22 and 8.78 and 4440 and 7330 μS/cm in Germencik. These values were recorded as 6.99–7.48, and on average 4500 μS/cm in Salavatlı; as 8.5–8.87, and on average 4974 μS/cm in Pamukören; and as 6.9–9.3 and 4120–5500 μS/cm in the Kızıldere region respectively. Along the GG, pH and EC values were measured as 6.8 and 2630 μS/cm in the Alaşehir region; and 6.14–7.60 and approximately 91.775 μS/cm in the Tuzla region (Table.1).

The results of chemical analysis and water types are presented in Table 2. Notable ion concentration variations can be seen in the BMG. Thermal waters have higher SO_4, lower Cl, boron ions in the east than the west in the BMG graben, and it is thought that this is due to the sea

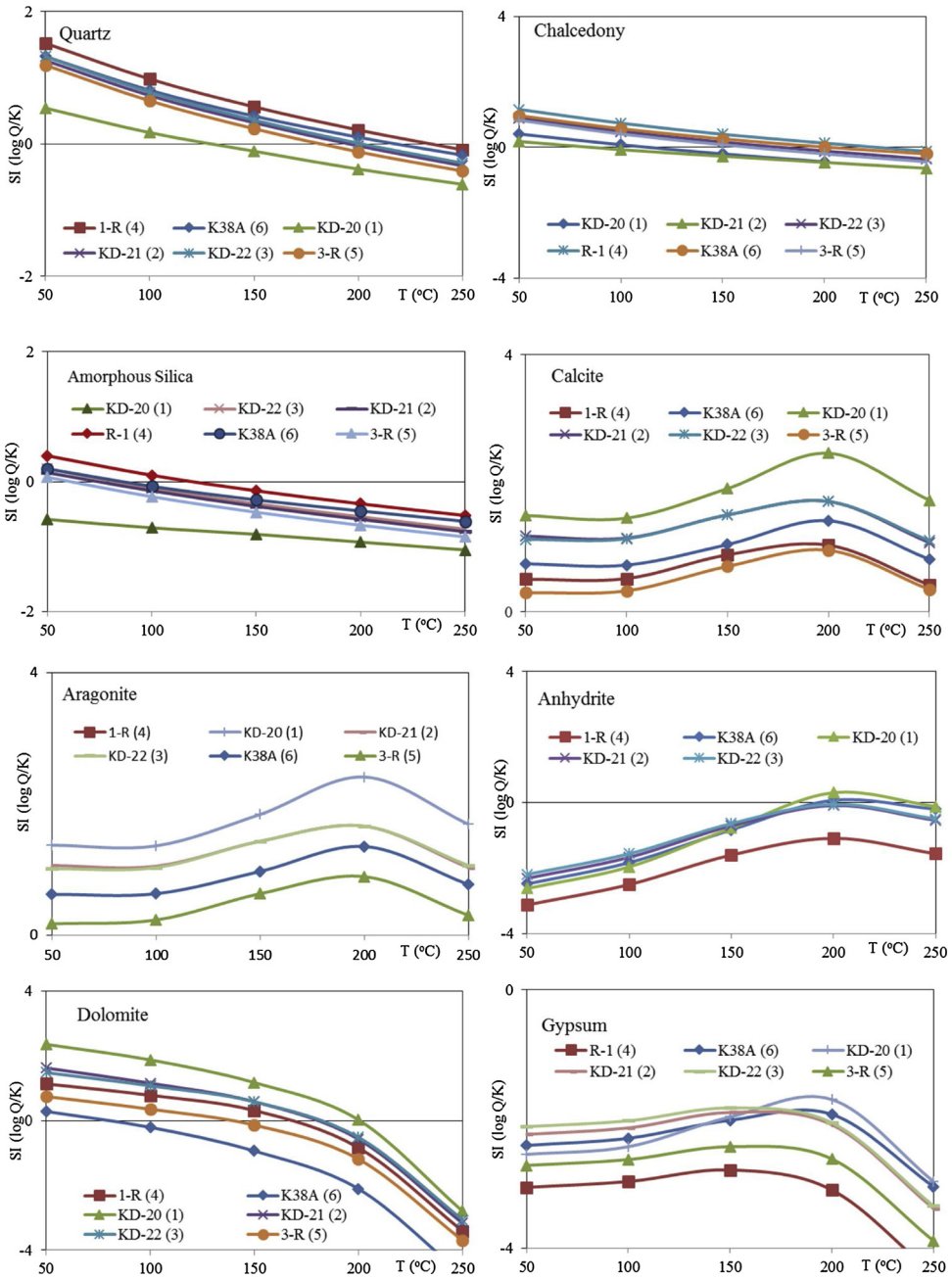


Fig. 12. Variation in the selected mineral saturation in thermal waters at different temperatures in the Kızıldere single flash type geothermal power plant. Numbers are the same as in Table 1.

intrusion effect in the western region of the graben (Haklıdır Tut et al., 2012). In the Gediz Graben, the boron concentration is remarkably high and the chemical composition of the thermal waters of Tuzla is different, with high Mg and extremely high Cl ion concentrations from other selected water samples, because of interaction between the sea and thermal waters in the region (Table 2). This type of water composition directly affects thermal water treatment, and it may cause many operational problems due to scale precipitation during power plant production.

There are limited open sources for NCG of thermal waters in Western Anatolia however; the main NCG concentrations in the region are given in Table 3. The dominant gas in both the BMG and GG systems is CO_2; the origin of CO_2 is due to the metamorphic rocks found in the geothermal reservoirs. (Haizlip Robinson et al., 2013). The second important gas component is H_2S, with its concentration highest at the western edge of the BMG and GG.

A general evaluation of hydrogeochemical characteristics of the selected geothermal fields have been has been carried out with the aid of Figs. 7 and 8. Fig. 7 shows major ion concentrations in thermal waters produced from selected geothermal fields. The Schoeller diagram shows common trends and gives an overall idea of absolute cation (Ca^{2+}, Na^{+} and Mg^{2+}) and anion (Cl^{-}, SO_4^{2-} and HCO_3^{-}) distribution in the waters as a semi logarithmic function of the concentration (meq/L). The thermal waters show similar trends and water types for each field. The water samples from the Tuzla geothermal field are very high in salinity (up to 30,000 mg/L of chloride) and exhibit a Na-Cl water type. In Germencik and Salavatlı are the concentrations of chloride (Cl^{-}) and bicarbonate (HCO_3^{-}) are similar. The Pamukören, Alaşehir and Kızıldere waters are Na-HCO_3 water types and have a relatively low salinity. The Kızıldere water is also high sulfate (SO_4^{-2}) water

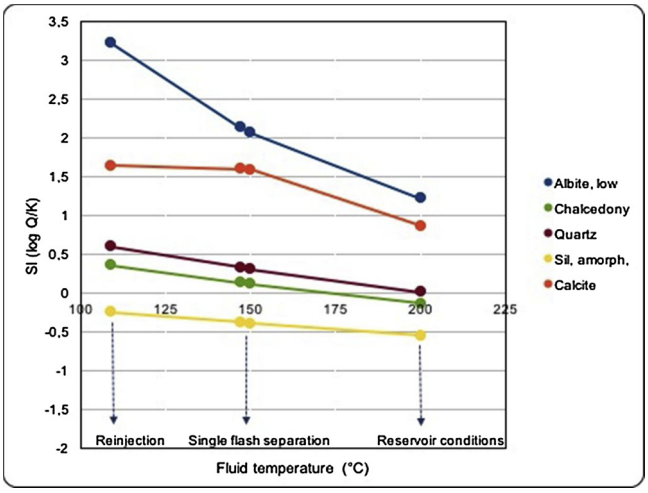


Fig. 13. Critical mineral precipitation tendencies from borehole to reinjection process for the Kızıldere single flash system in Western Anatolia.

compared to that from other fields and is of the Na-HCO_3-SO_4 water type.

The Piper diagram is the most convenient method, which shows different hydrochemical faces or origins of groundwater (Fig. 8). The Piper diagram for the groundwater in the selected area shows the existence of the following three main groups of waters: (1) Sodium bicarbonate waters are located mainly in the eastern part of the study area whose reservoir rocks consist of metamorphics (gneiss, schist and marble) and sediment rocks including limestone. Water samples No. 7, 8, and 9 of Kızıldere have high concentrations of sulfate. The source of sulfate may be attributed to the dissolution of minerals like gypsum-anhydrite ($CaSO_4$) of the sediment rocks (Tarcan et al., 2016). The other source of sulfate is traced to the oxidation of H_2S gas escaping from magma (Ozgur et al., 2004). (2) Sodium–bicarbonate–chloride waters are located in the Germencik and Salavatlı geothermal fields. Metamorphic rocks (marble, quartzite and mica schist) are reservoir rocks for their thermal waters. Cl^- is also derived from hot saline waters, which is derived from the faults (Baba and Sözbilir, 2012). (3) Sodium–chloride waters are characteristic for Tuzla region, which is located in the northwestern part of West Anatolia. Isotopic composition studies show that the Tuzla waters' origins are different from the others. The thermal waters are originated from deep-seated hot brine. The reservoir rocks consist of granitic rocks at a depth and also volcanic units at a depth of between 300–350 m (Baba et al., 2009).

In summary, the predominance of Na-HCO_3 and Na-HCO_3-SO_4 water types in Alaşehir, Kızıldere and Pamukören indicates the dissolution of alumina-silicates and gypsum-anhydrite ($CaSO_4$) of the sediment rocks. The thermal waters are dominated by mineral dissolution-saturation and ion exchange reactions. Na-Cl and Na-HCO_3-Cl water types in Tuzla, Salavatlı and Germencik are also related to hot brine at different depths.

Fig. 9 shows the Na-K-Mg geo-indicator proposed by Giggenbach, 1988. As shown in the Na/400–K/10–√Mg ternary diagram, thermal waters (except in Tuzla) mostly appear to have partially equilibrated with reservoir rocks. The geothermal fluids from the Tuzla field plot in a linear array with a full equilibration temperature of 160–220 °C. A high Mg concentration may suggest a water-rock interaction, dilution reactions, possibly affected by the absorption of dissolved reservoir rock. Some of the waters (24, 28, and 29) are plotted in immature waters. It could be explained by mixing with cold shallow groundwater and /or meteoric water. Most thermal waters in Germencik (11, 12, 25, 26, and 27) fall in the partially equilibrium and nearest Mg corner. This plot may result from interaction with the host rock such as the high Mg dolomite-limestone rocks. The rest are in partial equilibrium at temperatures around 200 °C in Salavatlı and Germencik, 270 in Alaşehir, 220-280 °C in Kızıldere, 200–260 °C in Pamukören geothermal fields.

3.3. Scale precipitations (SP) at flash, multi-flash and advanced geothermal power plants in WA

3.3.1. SP in single flash systems

Geothermal fluids consist of hot water, steam and gas in water-dominated systems. They can be used efficiently for power generation in flash type geothermal plants if the reservoir temperatures are near 200 °C or higher. In these systems, the water-steam mixture is separated into steam and water phases, with a minimum loss of pressure in a separator system (DiPippo, 2016). The separated steam is sent directly to the steam turbine or following purification by a scrubber or demister system to increase steam quality. Steam turbine and generator systems provide power generation in the system. Although scrubber and demister systems increase steam quality, a small portion of the steam is still wet because of moisture. This steam and non-condensable gases such as CO_2, H_2S etc. present the turbine is directed to a condenser system at low pressure. Water from a cooling tower is sprayed inside the condenser and steam is condensed and it is pumped to the cooling tower with all gases for heat rejection to the atmosphere.

A separated water phase (brine) is transferred to a reinjection well

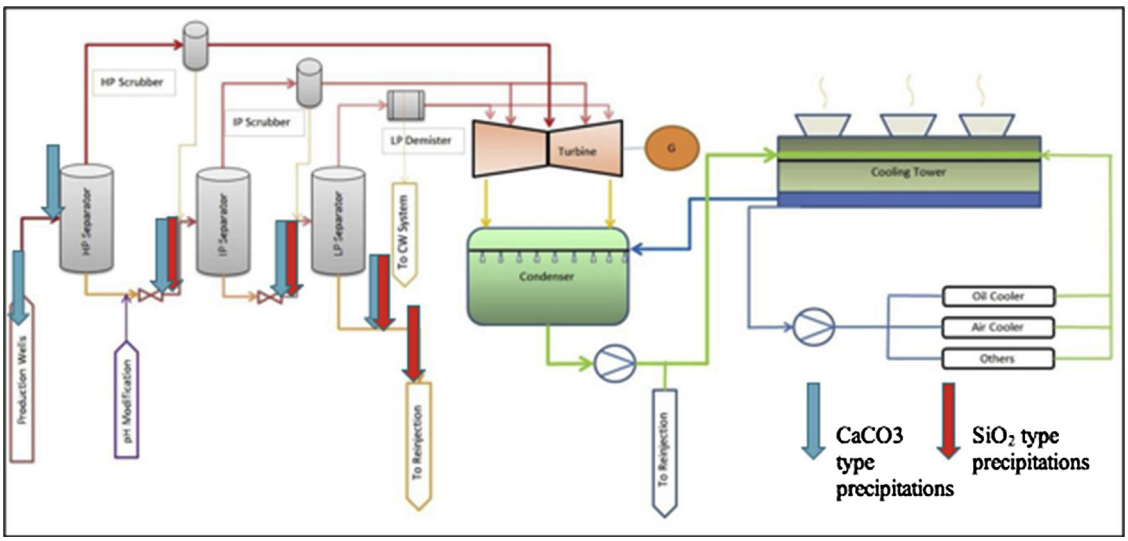


Fig. 14. Working principle of a multi-flash (triple flash) type geothermal power plant, with possible precipitation points in the system (Richardson et al., 2014).

Table 5
Scale types at Kızıldere-II the triple flash power plant (Haklıdır and Şengün, 2016). (Sampling Points:IP: Intermediate Pressure Separator, LP: Low Pressure Separator, HP: High Pressure Separator. 1-Before IP-LP Control Valve (CV), 2- After HP-IP CV, 4-Before HP-IP CV, 5- Before IP-LP CV, 6- On CV IP-LP, 7- After HP-IP CV).

Observed Scale Points	Na_2O %	MgO %	Al_2O_3 %	SiO_2 %	P_2O_5 %	SO_3 %	K_2O %	CaO %
1	0.077	14.55	2,068	27.78	0.104	0.0591	0.3826	11.41
2	< 0.014	6,012	1,182	12.46	0.7029	0.0328	0.1876	36.36
4	< 0.014	18.83	2,413	37.54	0.4684	0.0754	0.3394	2
5	< 0.014	10.79	1,626	21.65	0.2876	0.029	0.382	17.93
6	< 0.014	18.45	2.57	36.87	0.0204	< 0.0005	0.5908	1,805
7	< 0.014	18.12	2,569	36.58	0.0202	< 0.0005	0.6118	2.07



Fig. 15. Kızıldere-III triple flash + binary GPP (Unit-1; 96,5 MWe) on the BMG (Photo credit: Zorlu Energy).

by re-injection lines (Fig. 11). This is called a single-flash system because of only one separation of the water and steam phases in the system.

There is one installed single flash system, and it is located in Kızıldere (city of Denizli) in Western Anatolia (Fig. 11a). The installed capacity is 17.2 MWe, and it is the first geothermal power plant in Turkey. The plant is called Kızıldere-I and it was installed in 1984 by the Turkish government. The plant was privatized in 2008 by the government, following which, effective production started with the rehabilitation of the existing system (e.g., providing effective scale inhibition and reinjection applications) (Haklıdır Tut et al., 2010; Kındap and Haklıdır Tut, 2011). The scale precipitation and analysis of the scale samples are reported in Fig. 11b, c and Table 4.

After the decision was finalized to make the investment and install a new geothermal power plant, Kızıldere-I GPP was connected to Kızıldere-II GPP; some old wells were closed in Kızıldere-I GPP and a common re-injection strategy was started in the Kızıldere geothermal field in 2013 (Haklıdır and Şengün, 2016).

In this study, saturation indices were calculated at different temperatures, from 50 °C to 250 °C for the selected minerals by using chemical equilibrium modeling with PhreeqC software (Parkhurst and Appelo, 1999) for the selected wells of Kızıldere-I GPP such as KD-20, KD-21, KD-22, R-1, and R-3 samples from production wells, as well as the KD-38 A reinjection well in Fig. 9. Quartz and chalcedony show the same curve for the thermal waters. The fluids are generally supersaturated with respect to these minerals below 200 °C and below 135 °C, respectively. These thermal waters are supersaturated with respect to amorphous silica below 75 °C, except that of KD-20 which is undersaturated at all temperatures. The precipitation of silica is strongly dependent on the temperature and pH levels. Variation might be ascribed to the pH levels in the solution in single flash systems, as well as the temperature and pressure, which induce flashing, and (Brown, 2013; Utami et al., 2015).

The fluids are significantly supersaturated with respect to calcite and aragonite at every temperature in all wells in the Kızıldere geothermal system in the BMG. They are under-saturated with respect to dolomite between 150 °C and 170 °C (Fig. 12). The greater the amount of CO_2 dissolved in the geothermal water, the greater the amount of calcite that can be dissolved. Therefore, at the level of boiling in the production wells, calcite is precipitated. When the loss of CO_2 increases in the vapor phase, calcite will be deposited accompanying a pressure decrease (Brown, 2013; Tarcan et al., 2016). Anhydrite ($CaSO_4$) and gypsum ($CaSO_4.2H_2O$) are no scaling risk because the fluid is undersaturated with respect to both at all temperatures in all wells in Kızıldere.

In summary, there are different potential mineral precipitations from reservoir conditions to the end of the reinjection process in a single flash geothermal system (Fig. 13). Based on water-rock interaction at high temperature reservoir conditions, calcite and some metal silicates such as albite tend to precipitate in boreholes. During the steam separation process, the temperature of fluids decreases and different silica minerals tend to precipitate at the surface equipment. Due to temperature of reinjection water, amorphous silica starts to precipitate and it directly affects reinjection pumps and reinjection lines. In Fig. 13, the reinjection temperature is around 109 °C which is calculated to prevent silica mineral precipitation during the reinjection process. The second important issue is the adjustment of pH values to minimize silica scaling in reinjection water. It is around 9.4 for the selected Kızıldere single flash system in Western Anatolia.

3.3.2. SP in multi-flash and advanced (flash + binary) systems

If reservoir temperatures are high enough (based on existing power plants; if higher than 200 °C) geothermal fluids can be flashed twice or three times in a geothermal system (Haklıdır and Şengün, 2016).

The flashing process starts in the well and continues through different pressured separators on the surface for geothermal power plants in water-dominated systems. In multi-flash systems, there is more than one separator system, and separated high-pressure (HP) steam is fed to the high-pressure stages of the steam turbine. The liquid waste phase, coming from the HP separator, is flashed again in second (sometimes in the third) medium or low-pressure (MP or LP) separators. Pressure and temperature drops result in a different composition of scales in the system (Fig. 14; Table 5).

Kızıldere-II 80 MWe and Kızıldere-III 165 MWe (consisting of two units) were installed as triple flash + binary systems in the eastern region of the BMG (Fig. 15). 2 × 47.2 MWe double flash type power plants were also installed in Germencik (Aydın) in the western region of the BMG. There is one 45 MWe installed double flash + binary power plant in the town of Alaşehir (Manisa) in the GG.

In multi-flash systems such as Kızıldere-II and III, Germencik (Galip Hoca and Efeler-1 double flash power plants), geothermal water becomes supersaturated with respect to calcite as soon as steam begins to separate from the brine in a well. The degree of super-saturation with respect to calcite is greatest in a wellbore (Osborn et al., 2007; Haklıdır and Şengün, 2016; Stapleton et al., 2016). Silica scaling is also directly controlled by the re-injection temperature and the pH value following the production period for these types of power plants.

For multi-flash systems, saturation indices were calculated at different temperatures from 50 °C to 250 °C for the selected minerals using the chemical equilibrium modeling of PhreeqC software for selected wells of the Kızıldere-II triple flash + binary GPP such as: KD-18 A, KD-23B, KD-23D; wells belonging to the Germencik double flash GPP, such

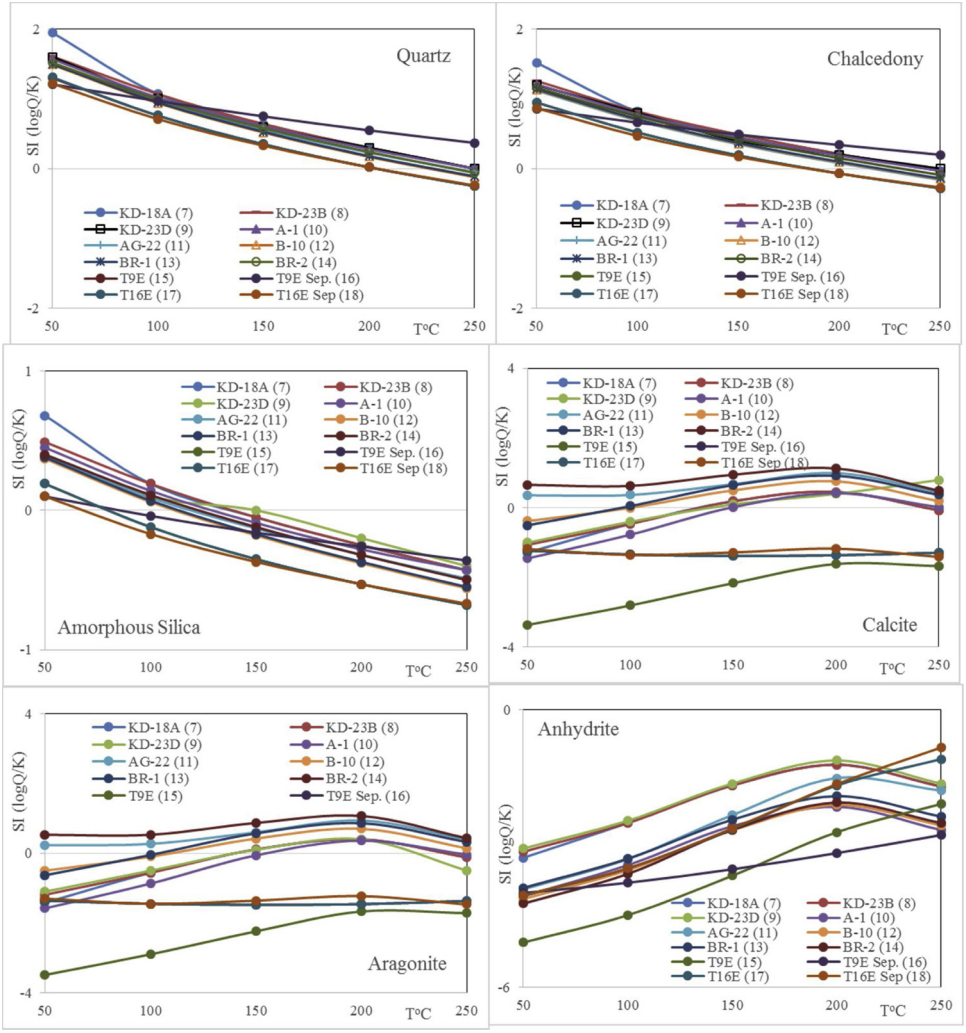


Fig. 16. Mineral saturation-temperature diagrams for thermal waters in the multi-flash and advanced (Flash + binary) system geothermal power plants. *Numbers are the same as in* Table 1.

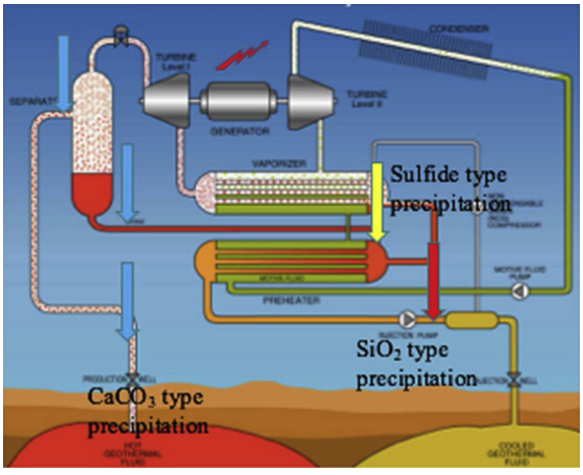


Fig. 17. The working principle of an ORC-binary geothermal power plant, showing possible mineral precipitation points in the system.

as AG-22, B-10, BR-1, BR-2 in the BMG; and the A1 well for the Alaşehir double flash + binary GPP within the GG (Fig. 16). In addition to data for these wells selected for SI calculations for multi-flash or advanced (flash + binary) systems results for, wells from one binary power plant (i.e. T9, T16) were used as benchmarks in the graphs.



Fig. 18. Binary geothermal power plant in Germencik (Aydın) in the BMG (Photo credit: ORMAT company).

The SI-temperature graphics for quartz and chalcedony are shown in the same curve in Fig. 16. The fluids are mostly supersaturated with respect to these minerals below 250 °C. Although, they supersaturated with respect to amorphous silica below 70 °C in Tuzla, and to the thermal water minerals below 140 °C in other fields.

The fluids are significantly under-saturated with respect to calcite and aragonite at each temperature in the benchmark Tuzla region wells. The thermal waters are supersaturated with respect to these minerals above 100 °C in the Germencik geothermal system. The other waters are supersaturated with respect to these minerals when temperature above

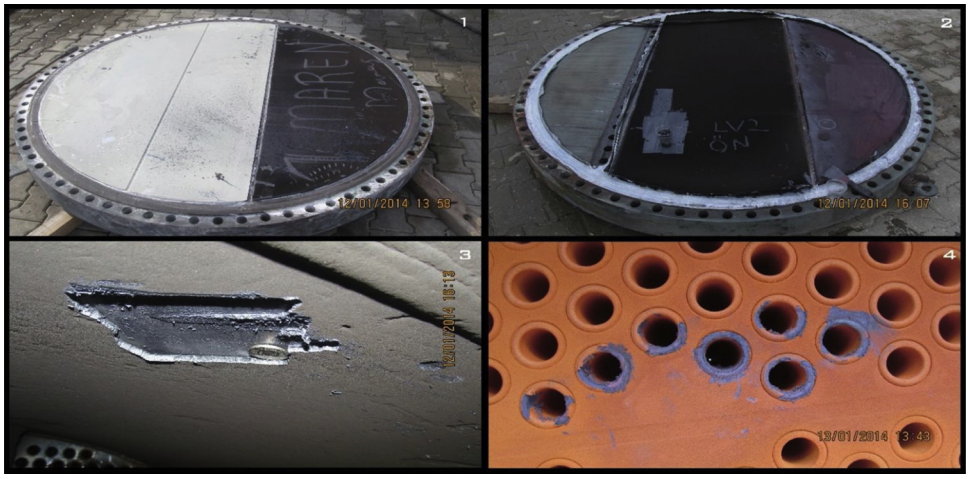


Fig. 19. Scales encountered at heat exchangers in the ORC-binary system in BMG; 1: High Pressure (HL) vaporizer, 2: Low Pressure (LP) vaporizer, 3: HP & LP vaporizer, 4: preheater last pass (Kaypakoğlu et al., 2015).

150 °C. The fluids are also under-saturated with respect to anhydrite at each temperature in all fields.

Controlling and evaluation of mineral precipitation tendencies are more complex in multi-flash systems than single flash systems. As an example; Kızıldere-II and III geothermal power plants in Western Anatolia were designed as multi-flash + binary cycle plants. The Kızıldere-II phase has three separation systems; a high pressure separation (HPS) system operating at 9.4 bara an intermediate pressure separation (IPS) at 4.72 bara and a low pressure separation (LPS) operating at 1.59 bara. It means that steam and water phases are separated at 3 stages to increase the system efficiency at high temperature reservoir conditions (around 235 °C). Depending on the number of separation systems, the ncg gas concentration decreases; they are mostly released with the first steam at HPS, and pH values of fluids tend to increase from HPS to reinjection.

Changes in fluid chemistry can be calculated from the analytical results for each separation stage considering P, T conditions. The chemical composition of the fluids can be calculated using the following formula from Henley et al., 1984;

$$C_{total} = C_{vapor} * y + C_{brine} * (1\text{-}y) \quad (1)$$

In formula (1), y is the steam fraction (Arnorsson, 1995; Verma, 1996) at the flashing pressure of each separator system. C_{total} is the concentration in a separator, C_v is the concentration in the vapor and C_{brine} is the concentration in the geothermal brine.

In Kızıldere-II, mineral precipitation tendencies are mainly observed as illustrated in Fig. 13. Due to higher reservoir temperature (as compared to Kızıldere-I), precipitation of calcite and the metal silicates can be seen. After the HP separator, there is greater tendency for silica minerals to precipitate (rows 6 and 7 in Table 5). However, high reinjection temperature (around 109 °C) prevents potential silica scaling in the surface equipment, reinjection pumps and reinjection pipelines. Calcite and metal silicates still tend to precipitate after the last stage of steam separation, but less so than in the higher temperatures parts of the system at Kızıldere-II GPP.

3.3.3. Scale precipitations in ORC-type binary geothermal power plants in WA

If the geothermal reservoir temperature is approximately 150 °C, binary cycle power plants realize a better performance (DiPippo, 2016). In an ORC binary cycle power plant, the thermal energy of the geothermal fluid is transferred through a heat exchanger from the geothermal fluid to the working fluid, which is then vaporized and sent to the turbine-generator system to produce energy (Fig. 17). The working fluid composition can be n-pentane, n-butane or a mixture of these organics, and it helps provide superheated steam for the system (Stapleton et al., 2016). ORC binary systems generally use air-cooled condensers, which may directly affect production performance in hot climate conditions or summer because of insufficient cooling in the system.

To increase the efficiency of an ORC power plant, one potential option is the geothermal-solar hybrid system (Bassetti et al., 2018). In a binary system, in addition to using a two-phase geothermal fluid, it is possible to use steam directly, as is done in the Kızıldere-II binary unit; this also reduces scale precipitation in the system (Richardson et al., 2014). The steam is from the HP steam separator for the Kızıldere-II binary unit in Western Anatolia.

In Western Anatolia, 34 ORC binary type geothermal power plants have been installed on the BMG and GG till 2018. Examples are: Pamukören (Aydın; 22,5 × 2 MWe), Salavatlı (Aydın; Dora I: 8,5 MWe, II:11,5 MWe, 3: 34 MWe, 4: 17MWe), Germencik (Maren 1: 20 MWe, 2:24 MWe, 3: 24 MWe) and Alaşehir-I (Manisa; 24 MWe) (EMRA, 2018; Fig. 18). Only a few of them are located in Tuzla (Çanakkale) and Afyon city, which are far from the large graben systems in Western Anatolia.

In binary type power plants, in addition to the calcite scaling in production wells used as flash cycle systems, the heat exchanger is critical in terms of scale precipitation (Haklıdır Tut and Balaban Özer, 2018). This is particularly the case regarding the precipitation of metal sulfide types such as stibnite, which may be observed at the vaporizer and preheater in the system (Fig. 19). Stibnite may be present due to gneiss and granite type rocks in geothermal reservoirs (Dost, 2018).

High temperature geothermal brines frequently contain antimony at low concentration, and arsenic at somewhat larger concentration. At ORC-binary power plants, increased H_2S concentration and decreased pH caused by returning steam or condensate to the brine may cause precipitation of stibnite in heat exchangers at lower temperature. Less commonly, arsenic sulfide, which is more soluble, may also precipitate under these conditions (Brown, 2013).

Stibnite dissolves in water to form the “hydroxide” $Sb(OH)_3$ as follows;

$$Sb_2S_3 + 6H_2O = 2Sb(OH)_3 + 3\ H_2S \quad (2)$$

However, in solutions where H_2S is present like in geothermal reservoirs, stibnite can also dissolve and then form sulfosalts of antimony;

$$Sb_2S_3 + H_2S = H_2Sb_2S_4 \quad (3)$$

Normally, $H_2Sb_2S_4$ may be a significant problem if the H_2S concentration is very high or the fluid temperature is relatively low with H_2S present (Brown, 2013).

Principally, it behaves like a silica mineral, and may precipitate

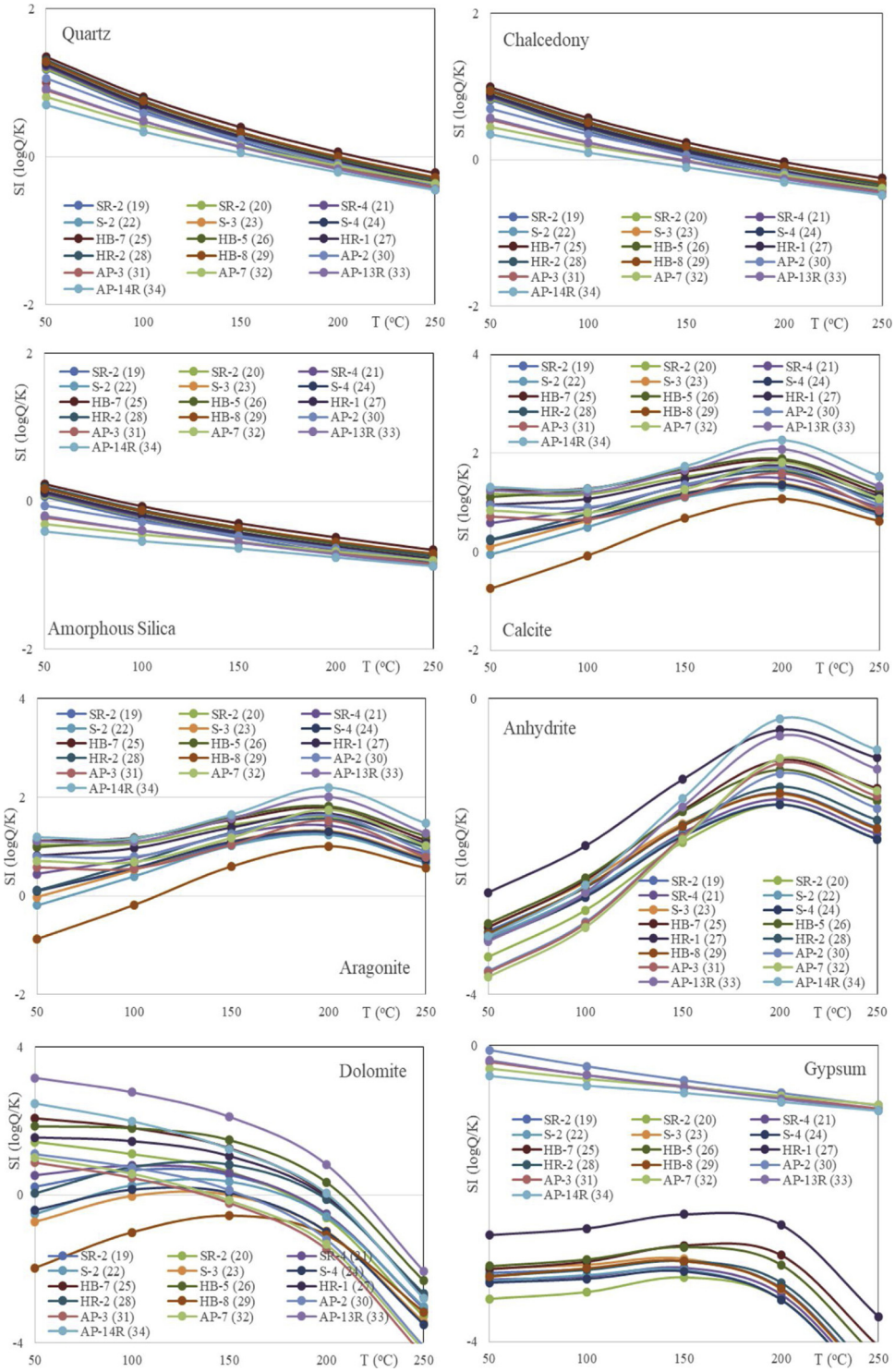


Fig. 20. Mineral saturation-temperature diagrams for thermal waters in binary system geothermal power plants. *Numbers are the same as in* Table 1.

when the brine temperature decreases in the system. Thus, during the production process between different pressure vaporizers or following the vaporizer, such as in the preheaters and possibly reinjection pumps, lines and wells may be subject to a stibnite precipitation risk (Dost, 2018).

For the binary systems, the saturation indices were calculated at different temperatures from 50 °C to 250 °C for the selected minerals using the chemical equilibrium model PhreeqC for selected well fluids in Salavatlı, Pamukören, and the second geothermal field in Germencik, where reservoir temperatures are less than 175 °C (Table 2).

In binary systems, thermal waters are supersaturated at temperatures below 160 °C with respect to minerals such as quartz and chalcedony (Fig. 20). Otherwise, the thermal waters of the Pamukören geothermal area are under-saturated with respect to amorphous silica at all temperatures. The fluids are under-saturated with respect to these minerals at approximately 70 °C in the Salavatlı and Germencik geothermal systems. The fluid is supersaturated with respect to calcite at all temperatures in most wells. Dolomite forms very slowly and other carbonate minerals such as calcite and aragonite are rapidly precipitated from the brine (Ford and Williams, 1989). The precipitation

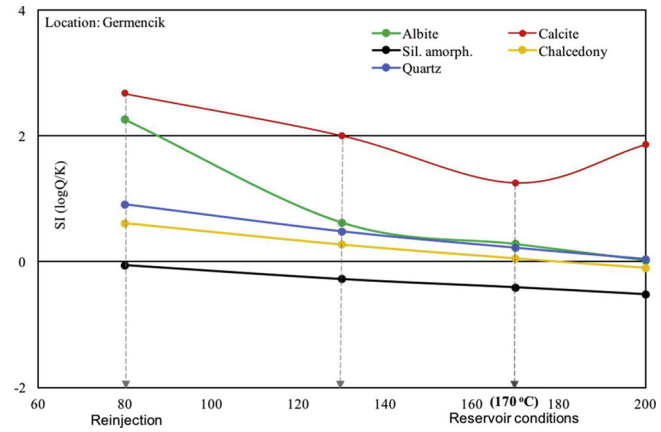


Fig. 21. Critical mineral precipitation tendencies from borehole to reinjection for the ORC-binary system in Western Anatolia.

tendency of dolomite starts below 100–150 °C in Salavatlı and somewhere below 140–175 °C in Germencik and Pamukören. Thermal waters are under-saturated with respect to gypsum at all temperatures.

For the selected ORC-binary type power plant for this study, calcite may precipitate in the bore hole and throughout the system (Fig. 21). As the fluid flows from the reservoir and through the system, metals silicates and amorphous silica may also precipitate with decreasing temperature. At binary type power plants receiving fluids from the medium temperature geothermal systems in Western Anatolia, reinjection temperature is lower than with flash systems and spent brines require treatment before they are reinjected. While stibnite is a potential precipitant at ORC-binary systems, super-saturation in respect to stibnite is not encountered at the temperatures considered in the power plant selected for this study. More sensitive analysis of Sb at the low concentrations encountered may help to better understand behavior of Sb at Germencik.

4. Control of mineral precipitation and effective scale inhibition methods in geothermal power plants in WA

In geothermal systems, physical and chemical changes start with CO_2 degassing and steam flashing in a borehole. Mineral concentrations and pH increase at that flash point, with microcrystalline $CaCO_3$ and other metal carbonates as well as metal silicates precipitating on casing walls. The second flash point occurs after the flow control valve, which also serves as a wellhead pressure reducer to decrease the line pressure. Flashing for steam generation by using different pressure separator systems begins after these steps. Up to this point, depending on water chemistry, the precipitation processes for single/multi-flash and binary systems are similar. Among all geothermal power cycles, the path of the fluids is difficult to follow due to multiple separations in multi-flash systems. However, the efficient operation of a geothermal power plant depends on mitigated or controlled scale precipitation (Jamero et al., 2018).

The control of precipitation in a borehole and before/after the separator system is less challenging for single flash and binary systems than for multi-flash systems (Figs. 10, 17). Multi-flash systems require stricter control of deposits in the borehole, at each stage of steam separation (HP, IP and LP) and in the reinjection lines, because of the large drops of temperature and pressure as the brine flows through the system. As an example, the Kızıldere triple flash system has nine production wells and three pressure stages (HP = 9,3 bara, IP = 4,2 bara and LP = 1,83 bara). The average reservoir temperature is approximately 220 °C for this geothermal field. In this system, between the wellhead and separator, a control valve reduces the pressure in the pipe. The temperature of the fluids drops to approx. 175 °C before HPS. All geothermal fluids collected from the nine wells are mixed, and the fluid mixture is transferred to the HP separator systems. The residual brine is sent to the IP separator systems, where further expansion takes place to 4,2 bara and 145 °C.

Subsequently, the last brine is sent to the LP separators, from which it leaves at a pressure of 1,83 bar and 109 °C before reinjection (Haklıdır and Şengün, 2016). The reinjection temperature is also critical to prevent silica scaling, with the lowering of temperatures occurring after leaving the separator system. Silica scaling creates challenges along reinjection lines; it is possible to reduce the problem through the addition of condensed steam to the brine phase prior to entering the preheater (Sigfusson and Gunnarsson, 2011; Richardson et al., 2014;)

Due to the pressure reduction and the release of steam and CO_2, the pH and the ion concentrations gradually increase; while the pH of brine is approximately 7 after the control valve (wellhead), it is around 7.5 after the HP separators, 9.4 after the IP- and ≥ 9.6 after the LP-separators. Due to pressure drops based on a CO_2 reduction in the system, the pH and the ion concentrations begin to rise; while the pH of brine is around 7 after the control valve at the wellhead, it is around 7.5 after the HP separators, 9.4 after the IP- and ≥ 9.6 after the LP-separators (Fig. 14). All these processes result in calcite, carbonate and silicate minerals or sometime metal sulfides accumulating along the system.

In ORC-binary systems, reservoir temperatures are lower than those of the flash temperatures and additional heat is provided by a second organic working fluids. In this case, the heat exchanger system is a critical second point in terms of mineral precipitation after a borehole. Especially high and low pressure vaporizers of heat exchanger systems (Fig. 19) are critical locations for the precipitation and then besides



Fig. 22. Inhibitor application for a production well for single flash and binary type power plants (Photo from: Kızıldere-II GPP; PW: Production well).

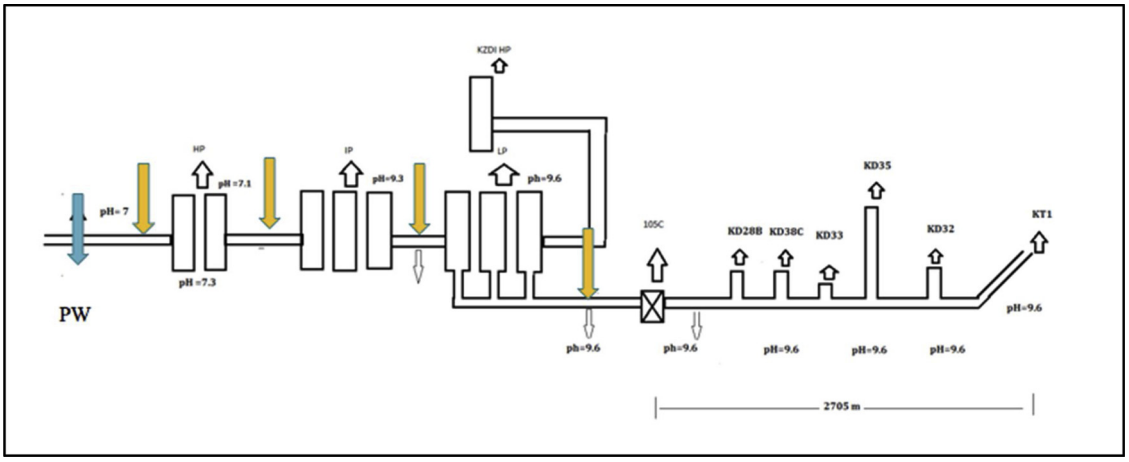


Fig. 23. Inhibitor application for the multi-flash systems (Basic design belongs to Kızıldere-II triple flash GPP).

calcite precipitation, stibnite mineral starts to precipitate, depending on temperature decreases in the system. Due to low reservoir temperatures, silica precipitation is not a major concern for binary systems.

4.1. Effective scale inhibition methods for geothermal power plants in WA

Prevention of scaling has been addressed by chemical inhibitors and adjustment of temperatures since 2009 in Western Anatolia. Prior to this period, there were no effective inhibitor applications, and mechanical cleaning was preferred by the field owner in the Kızıldere-I GPP.

Scale inhibition applications in geothermal power systems have been applied using phosphate salts and phosphonate salts + polymer mixture type chemical inhibitors in WA. Phosphonic acids or phosphonates are organically based compounds containing one or more C-PO $(OH)_2$ or CPO $(OR)_2$ (R = alkyl, aryl) groups (Kılıççıoğlu, 2009). These types of inhibitors are highly soluble in water and phosphonate scale prevention inhibitors-Diethylenetriamine penta (methylene phosphoric acid) (DETPMP; Baraka-Lokmane and Sorbee, 2010)- are known to be very effective for carbonate type scales. Phosphonate salts are identified as the result of deprotonation of phosphonic acids (Savignac and Iorga, 2003).

Although phosphonate type chemicals are very effective for calcite inhibition, some phosphonate acid type chemicals may corrode metal borehole equipment (such as capillary tubing and diffusers).

Long term inhibitor application tests show that if reservoir temperatures are lower than 200 °C, phosphate type inhibitors can be effectively used for the production phase in geothermal systems. At above 200 °C, phosphonate type chemicals may lose stability, which leads to the use of polymer + phosphonate types in combination, particularly in multi-flash systems such as those at Kızıldere-II, III and Germencik (Galip Hoca and Efeler I GPPs) along the BMG. It has been observed that phosphonate gel forms in tubing when the temperature rises above 200 °C (Haklıdır and Şengün, 2016). Although the reservoir geology at BMG and GG is similar, it is important to note that the same inhibitors cannot always provide the same positive results, most probably because of a different gas composition and different concentrations of some ions in the water phase in geothermal power plants in Western Anatolia.

To prevent scale precipitation in a production well, the inhibitor must be injected into the flowing well through tubing at 10–50 m below the estimated gas-breakout depth. In a multi-flash system, inhibitor dosage systems must be installed ahead of pressure drop points to protect the mechanical equipment also (Fig. 22).

It is noteworthy that decisions on a suitable inhibitor via field tests are quite important. After selection of the most suitable inhibitor for the system, a correct dosing of the inhibitor solution is also critical. The sustained inhibitor application directly depends on the dosages of chemicals and the dosages for each well can be determined by short-term field tests. While under-dosage application compared to the optimum value poses a risk to the system, over dosage is also problematic, causing different mineral precipitations such as of calcium phosphate in the system (Haklıdır Tut and Haklıdır, 2017).

Inhibitor dosing is required at the wellhead and before the separator system for single flash systems, if the plant operator pays attention to the reinjection temperature of the brine. For multi-flash systems, inhibitor dosing is required for the wellhead, prior to the HP separator system, between different pressure separator systems, and following the IP separator system (Fig. 23).

Stibnite precipitation has been observed in the Pamukören and Germencik (Efeler-4) binary power plants (Kaypakoğlu et al., 2015; Stapleton et al., 2016; Fig. 19). In binary type power plants, stibnite precipitation starts after the vaporizer, preheaters, or reinjection phase, depending on rapid temperature and pressure changes. Although scale prevention inhibitor companies' research and development studies are still continuing on stibnite precipitation; pH adjustment, cleaning of equipment with high pressure water jets or the use of caustic chemicals have also been tested options for dissolution of the mineral in ORC-binary power plants in Western Anatolia. It should be noted that these methods have side effects such as calcite precipitation, and corrosion. (Kaypakoğlu et al., 2015). The most effective method appears to be the dosing of a suitable inhibitor immediately prior to the preheater in binary power plants.

5. Conclusion

Turkey has an important geothermal power source as well as different geothermal power technologies producing energy in Western Anatolia. These geothermal power plants have high capacity factors and they can only be sustainable and cost effective if reservoirs and power plants operate efficiently during long license durations.

There are a few critical chemistry challenges to be considered during design and operating phases of geothermal power plants. Depending on reservoir temperatures, fluid chemistry, and non-condensable gas composition and concentrations and fluid production rates, geothermal power plant operations require different maintenance applications such as prevention of scaling, corrosion, reservoir cooling problems because of water interactions, sea intrusion effects and the decline of reservoir pressure in geothermal reservoirs in Western Anatolia.

The most important challenge directly affecting steam production is mineral precipitation as different thermodynamic cycles require different inhibition applications. Calcite, other carbonates, metal silicates,

and metal sulfides can be found in the different parts of a geothermal system. In particular, changing pressure and temperature conditions control mineral precipitation from production to reinjection. The path of geothermal fluids and steam separation processes are more complex in multi-flash and advanced geothermal power plants which consist of flash + binary power plants than in other geothermal power systems. During the steam production process, each pressure drop point may induce an additional mineral precipitation in the system. The systematic physical and chemical monitoring studies are highly critical to provide an uninterrupted energy production required power plants.

Selection of suitable chemical inhibitors such as phosphonate, polymer based or mixed solutions, identifying optimum dosages of inhibitors along with correct dosage points, and preparing correct dilutions with water for each geothermal system (sometimes for each well) can extend the lifetime of geothermal power plant wells and equipment.

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